AI MODEL HEALTH MONITORING & AUTOMATED MAINTENANCE SYSTEM

Product Requirements Document

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1. Problem

Models in production slowly get worse. Teams find out late, guess the cause, retrain everything on a schedule, and sometimes ship a model that is worse than the old one.

2. Goal

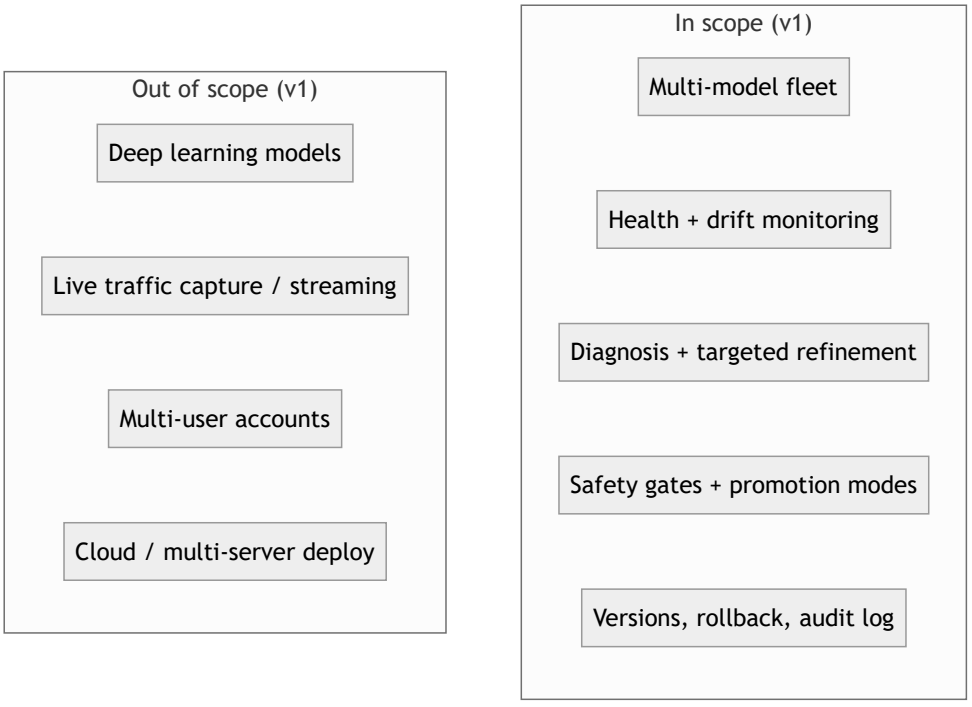
Give a small ML team one place to **watch** models, **understand** why they degrade, **fix** them safely, and **undo** any change.

Success looks like - A health problem is flagged within one monitoring run. - Every alert comes with a plain-English cause and a suggested fix. - No candidate reaches production without passing all 7 gates. - Any version can be restored in one click.

3. Users

User	Wants
ML engineer	Register models, run checks, review candidates
Ops / on-call	See fleet health and alerts at a glance
Reviewer / manager	Approve promotions, read the audit trail

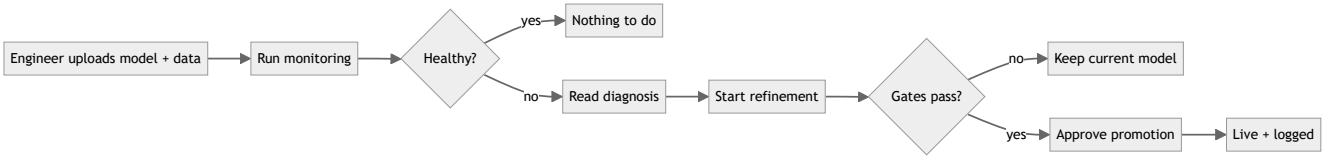
4. Scope



5. Requirements

ID	Requirement	Priority	Status
R1	Register many models, each isolated	Must	Done
R2	Upload models (sklearn, XGBoost, LightGBM, CatBoost) and datasets	Must	Done
R3	Health score from 5 pillars; honest "labels unavailable" state	Must	Done
R4	Tell expected drift (approved baseline) from unexpected drift	Must	Done
R5	Diagnose cause; pick one targeted fix	Must	Done
R6	7 safety gates before any promotion	Must	Done
R7	Modes: AUTO, ASSISTED (default), MANUAL	Must	Done
R8	Atomic rollback; full audit log	Must	Done
R9	Alerts with acknowledge / resolve	Should	Done
R10	Regression, multiclass, binary tasks	Should	Done
R11	Ranking metrics	Could	Example only
R12	User accounts and roles	Should	Not started
R13	Research data export (CSV)	Could	Script done, uncommitted

6. Main user story



7. Risks

Risk	Answer
Malicious model file runs code	Extension whitelist, hash check, child-process first load (see Security doc)
Bad model auto-deployed	ASSISTED by default; 7 gates; rollback
Noisy alerts	Expected-drift baselines
Outgrowing SQLite	Out of scope for v1